MILITARY SPECIFICATION
AIRBORNE DATA ANNOTATION SYSTEM AN/AYA-10( )

1. SCOPE

1.1 This specification covers one type of Airborne Data Annotation System (ADAS) designated AN/AYA-10( ) which automatically converts dc analog, synchro analog, and binary input data into numeric and excess-three binary coded decimal (+3 BCD) formats for display on recording head assemblies (RHA) of surveillance sensors. (See 6.1 and 6.4.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-handling, Wood (General Construction Requirements)
QQ-S-781	Strapping Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Non-metallic (and Connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-P-116	Preservation, Methods of
MIL-B-5087	Bonding, Electrical and Lightning Protection For Aerospace Systems
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-I-6181D (Cancelled)	Interference Control Requirements For Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish For Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40 Inch X 48 Inch Pallets. Skids, Runners or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Defects
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-782	Reconnaissance/Mapping Data Marking
MIL-STD-810	Environmental Test Methods
MIL-STD-826 (Cancelled)	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment and Facilities

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid, or request for proposal, shall apply.

Title 10, Part 20, Code of Federal Regulations (Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D.C.)

3. REQUIREMENTS

3.1 Description. The AN/AYA-10( ) shall consist of the following components:

Control Display Monitor (CDM) C-8338( )/AYA-10
Signal Data Converter (SDC) CV-2647( )/AYA-10
Base, Shock Mount, Electrical Equipment MT-4539( )/AYA-10
Battery BA-( ) (See 3.12)

3.1.1 Associated equipments. The AN/AYA-10( ) shall be capable of operating in conjunction with the avionic equipments listed below. The use of the simulated signals provided in the bench test set-up of 4.7.2.1 in the satisfactory performance of the tests specified herein shall satisfy this requirement.

Side-Looking Airborne Radar (SLAR) AN/APS-94D
Infrared (IR) Surveillance System AN/AAS-24( )
Inertial Navigation Set (INS) AN/ASN-86( )
Radar Altimeter AN/APN-171( )
Barometric Altitude Transducer TR-253( )/AYA
Camera System (FWD) KA-60( )
Camera System (AFT) KA-60( )
Camera System KA-76( )

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.1.

3.3 Procurement model. A procurement model of the AN/AYA-10( ) system will be loaned to the contractor. The procurement model shall be used as the basis for the physical, mechanical, electronic and electrical circuitry construction only of the equipment with exceptions as specified in the invitation for bids and the contract and as modified by the performance and

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construction requirements specified herein. Performance characteristics of the equipment shall conform to those specified herein and not to those of the procurement model.

3.4 Parts, materials and processes; general. In addition to the requirements of this specification, the selection requirements of MIL-E-5400 shall apply. (See 4.4.)

3.5 Time totalizing meter. The time totalizing meter shall be in accordance with MIL-M-7793 and shall indicate the total cumulative time the SDC is energized. (See 4.20.)

3.6 Finish, protective. Equipment shall be given protective finishes in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surface shall be film B in accordance with MIL-F-14072. (See 4.4.)

3.7 Marking (See 4.4.)

3.7.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.7.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial numbers. All items listed in 3.1 except the battery shall have serial numbers.

3.8 Preconditioning. The equipment shall be capable of meeting the requirements herein after subjection to the preconditioning of 4.6.

3.9 General characteristics.

3.9.1 Operation. The ADAS shall accept the inputs from the sensors listed in 3.1.1 and process this data to provide outputs to the film recording heads (RHA) of each sensor (IR, SLAR and camera systems) for annotation of the film in accordance with MIL-STD-782( ) with the exceptions noted in Table II. In addition, the ADAS shall accept fixed input information as defined in this specification, to be used to identify the mission. (See 4.7.2.1.)

3.9.2 Vg/H output and Cycling Rate.

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3.9.2.1 Vg/H output. A Vg/H signal shall be provided as a dc analog output voltage to all sensor systems except SLAR. The analog voltage shall vary from 0.5 Vdc to 100.6 Vdc representing Vg/H ratios of 0.01 to 2.00 for the KA-60( ) FWD and KA-76( ) camera systems and the AN/AAS-24( ) IR and from 0.5 Vdc to 50.3 Vdc representing Vg/H ratios of 0.01 to 2.00 for the KA-60( ) AFT camera system. When working into an output resistive load of not less than 6.8K ohms, the output Vg/H analog voltages shall maintain linearity with the Vg/H ratio inputs within 1% and shall correspond within ±5% to the nominal values listed in 4.10.5.1 and 4.10.6.1. Linearity shall be determined from a plot of Vg/H analog voltage outputs against the Vg/H ratio inputs. The best fit straight line through the data points shall not be displaced from the theoretical data line by more than 1% of the value of the theoretical data at any point. (See 4.10.5.1 and 4.10.6.1.)

3.9.2.2 Cycling Rate. The AN/AYA-10( ) system shall accept Cycling Rate input signals from the KA-60( ) FWD and KA-60( ) AFT camera systems for processing a maximum of six cycles per second and conversion to a form for presentation in the NUMERIC format indicated in Table II, Section E. The input signal voltages and corresponding Cycling Rates shall be as defined in 4.10.5.2 and 4.10.6.2.

3.9.3 Command navigation data request output. A data request signal shall be provided to the INS by means of a three wire code to identify the desired parameter. (See 4.10.12.)

3.9.4 Input data. The ADAS shall be capable of receiving input data signals of fixed flight and sensor data, variable navigational data and variable sensor data. The fixed flight and sensor data shall be manually programmed at the front panel of the CDM. The variable navigational data shall be obtained from the INS and Barometric Altitude Transducer, TR-253/AYA. In addition, the ADAS shall respond to data demands from all five sensors (3 camera systems, plus IR or SLAR) and a data ready signal from the INS. The input data shall be in accordance with Table I. (See 4.9, 4.10, 4.11 and 4.12.)

Table I. Input Data

Parameter	Code	Bits/Word	Signal Lines	Waveshape	Level	Toler
Northings	+3 BCD	24 parallel	24	Pulse	5V	±0.5V
Eastings	+3 BCD	24 parallel	24	Pulse	5V	±0.5V
Heading	+3 BCD	16 parallel	24	Pulse	5V	±0.5V
Radar Alt.	+3 BCD	20 parallel	24	Pulse	5V	±0.5V
Drift Angle	+3 BCD	16 parallel	24	Pulse	5V	±0.5V

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Table I. Input Data (Cont.)

Parameter	Code	Bits/Word	Signal Lines	Waveshape	Level	Toler
Gnd Speed	+3 BCD	16 parallel	24	Pulse	5V	±0.5V
Vg/H	+3 BCD	12 parallel	24	Pulse	5V	±0.5V
SLAR Range			3	DC Level	Open/28V	±2.0V
SLAR Range Delay			7	DC Level	Open/28V	±2.0V
KA-76 Angle			4	DC Level	0/28V	±2.0V
IR Filter			5	DC Level	0/28V	±2.0V
Data Demand				Pulse		
KA-76			1	20 millisec min	28V	±2.0V
IR			1	38 millisec min	28V	±2.0V
SLAR			1	38 millisec min	28V	±2.0V
KA-60(FWD)			1	500 usec ±250	6V	±1.5V
KA-60(AFT)			1	500 usec ±250	6V	±1.5V
Data Ready		(7 pulses per 18.5 millisec)	1	12 usec ±.5	5V	±0.5V
Cycling Rate						
KA-60(FWD)	*(31.14V per cycle)		1	DC Level	.503 to 100.6 Vdc	
KA-60(AFT)	*(8.93V per cycle)		1	DC Level	.503 to 50.3 Vdc	
Pitch			3	400 Hz	11.8 Vac max between lines	
Roll			3	400 Hz	11.8 Vac max between lines	
Synchro ref for Pitch and Roll			1	400 Hz	26 Vac	±2.5V
Barometric Altitude			1	DC Level	0 to 6 Vdc	

*Internal processing levels (see 4.10.5.2 and 4.10.6.2)

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Table II. Outputs Data To Be Annotated

A. The following is the data in numeric and BCD format to be annotated in each sensor. When recording position in relative coordinates, the following sign convention shall be used:

(1) Eastings - (UTM)

+ to the East
- to the West

(2) Northings - (UTM)

+ to the North
- to the South

B. Infrared Sensor (IR) -

NOTE: It should be noted that the IR and SLAR will not be used in the same airplane at the same time. Hence, these equipments will share the same annotation circuitry.

(1) Numeric -

Northing 0000.0 to 9999.9 KM (UTM and sign)
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99
Heading 000.0 to 359.9 degrees
Radar Altitude 0000 to 9999 feet
Ground Speed 000 to 999 knots
Frame Number 0000 to 9999
Time (Hours, minutes, seconds - 000000 to 395959)
Date (Day, month, year - 000000 to 391999)
IR Filter Number Identification 1 to 5
Sortie Number (0000 to 9999)
Taking Unit (000000 to 999999)

(2) BCD -

Northing 0000.0 to 9999.9 KM (UTM and sign) 1/ 2/
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99 1/ 2/
Drift Angle +99.9 to -99.9 degrees 2/
Radar Altitude 0000 to 9999 feet
Heading 000.0 to 359.9 degrees 2/
Roll -30.0 degrees to +30.0 degrees 2/
Pitch -30.0 degrees to +30.0 degrees 2/
Barometric Altitude 0000 to 9950 feet
Time 000000 to 395959
Date 000000 to 391999

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Table II. Outputs Data To Be Annotated (Cont.)

IF Filter Number Identification 1 to 5
Sortie Number 0000 to 9999
Taking Unit 000000 to 999999
Sensor Identification 110

C. Photographic Sensor KA-76( ) -

(1) Numeric -

Northing 0000.0 to 9999.9 KM (UTM and sign)
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99
Heading 000.0 to 359.9 degrees
Radar Altitude 0000 to 9999 feet
Ground Speed 000 to 999 knots
Frame Number 0000 to 9999
Camera Angular Position 1 to 5
Time 000000 to 395959
Date 000000 to 391999
Focal Length 00 to 99
Sortie Number 0000 to 9999
Taking Unit 000000 to 999999
Roll -30 degrees to +30 degrees
Pitch -9 degrees to +9 degrees

(2) BCD -

Northing 0000.0 to 9999.9 KM (UTM and sign) 1/2/
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99 1/ 2/
Drift Angle +99.9 to -99.9 degrees 2/
Radar Altitude 0000 to 9999 feet
Heading 000.0 to 359.9 degrees 2/
Roll -30.0 degrees to +30.0 degrees 2/
Pitch -30.0 degrees to +30.0 degrees 2/
Barometric Altitude 0000 to 9950 feet
Time 000000 to 395959
Date 000000 to 391999
Sortie Number 0000 to 9999
Taking Unit 000000 to 999999
Sensor Identification 090 to 109

D. Side Looking Radar (SLAR) -

NOTE: IR and SLAR will not be used in the same airplane at the same time. These equipments share the same annotation circuitry.

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Table II. Outputs Data To Be Annotated (Cont.)

(1) Numeric -

Northing 0000.0 to 9999.9 KM (UTM and sign)
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99
Heading 000.0 to 359.9 degrees
Radar Altitude 0000 to 9999 feet
Ground Speed 000 to 999 knots
Range 25, 50, 99 km
Range Delay 00, 10, 20, 30, 40, 50 and 60
Time 000000 to 395959
Date 000000 to 391999
Sortie Number 0000 to 9999
Taking Unit 000000 to 999999

(2) BCD -

Northing 0000.0 to 9999.9 KM (UTM and sign) 1/ 2/
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99 1/ 2/
Range Delay 1 to 7 KM
Heading 000.0 to 359.9 degrees 2/
Radar Altitude 0000 to 9999 feet
Barometric Altitude 0000 to 9950 feet
Roll -30.0 degrees to +30.0 degrees 2/
Pitch -30.0 degrees to +30.0 degrees 2/
Drift Angle +99.9 to -99.9 degrees 2/
Time 000000 to 395959
Date 000000 to 391999
Sortie Number 0000 to 9999
Taking Unit 000000 to 999999
Sensor Identification 111 to 113

E. Camera KA-60( ). (FWD and AFT) The FWD and AFT KA-60( ) Cameras have separate and independent data blocks.

(1) Numeric.

Northing 0000.0 to 9999.9 KM (UTM and sign)
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99
Heading 000.0 to 359.9 degrees
Radar Altitude 0000 to 9999 feet
Ground Speed 000 to 999 knots
Frame Number 0000 to 9999
Time 000000 to 395959
Date 000000 to 391999
Sortie Number 0000 to 9999.

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Table II. Outputs Data To Be Annotated (Cont.)

Taking Unit 000000 to 999999
Roll -30 degrees to +30 degrees
Pitch -9 degrees to +9 degrees
Cycling Rate 1 to 6 (AFT)
Cycling Rate 1 to 3 (FWD)

(2) BCD -

Northing 0000.0 to 9999.9 KM (UTM and sign) 1/ 2/
Easting 000.0 to 999.9 KM (UTM) and Zone Number 00 to 99 1/ 2/
Drift Angle +99.9 to -99.9 degrees 2/
Radar Altitude 0000 to 9999 feet
Barometric Altitude 0000 to 9950 feet
Heading 000.0 to 359.9 degrees 2/
Roll -30.0 degrees to +30.0 degrees 2/
Pitch -30.0 degrees to +30.0 degrees 2/
Time 000000 to 395959
Date 000000 to 391999
Sortie Number 0000 to 9999
Taking Unit 000000 to 999999
Sensor Identification 088 aft
Sensor Identification 089 forward

NOTES:

1/ Not contained in MIL-STD-782 format.

2/ BCD does not display a decimal point.

3.10 Self test. The status of self-test of the AN/AYA-10( ) system shall be indicated by the CRT display and the indicator light on the CDM. (See 4.27.)

3.11 CDM CRT display intensity. The intensity control circuitry shall provide for optimum intensity of the BCD and numeric displays and shall provide a continuous intensity of display from an extinguished beam to an oversized defocused dot. (See 4.26.)

3.12 Battery. The CDM shall employ a battery, assigned by the contracting officer, to provide a stable voltage for the frequency control circuit.

3.13 Reliability. The contractor shall perform the reliability evaluation tests in accordance with 4.23 and 4.23.1. AN/AYA-10( ) systems that are subjected to Group C tests shall not be subjected to reliability testing.

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3.14 Burn-In. The contractor shall perform burn-in in accordance with 4.22 on each AN/AYA-10( ) system prior to Groups A and B testing.

3.15 Systems safety engineering. The equipment shall be designed both electrically and mechanically in accordance with the safety requirements of MIL-STD-454, Requirement 1 and the safety requirements for personnel and equipment of MIL-STD-1472. In particular, the following requirements shall be met: (See 4.24.1.)

a. The system shall incorporate features to protect operators from contact with voltages in excess of 30 volts rms or dc and maintenance personnel from contact with voltages in excess of 70 volts rms or dc.

b. All contacts and terminals having voltages between 70 and 500 volts rms or dc must be guarded and clearly labeled, or permanently marked, to indicate highest voltage encountered upon removal.

c. All contacts and terminals having voltages over 500 volts rms or dc will be completely enclosed. The enclosure shall be labeled to indicate highest voltage encountered upon removal.

d. All external parts shall be of ground potential at all times during operation.

e. No parts of the frame or enclosure of electrical equipment shall be used as a current carrying conductor.

f. Intentionally added radioactive material will not be used unless there is no practicable substitute. If intentionally added radioactive material is used in any chemical/physical form (e.g. in self-luminous dials, electron tubes, thorium lenses, etc.), the equipment will be designed to insure that the radiation dose to individuals and the contamination of the environment will be no greater than the permissible limits prescribed in Title 10, Part 20, Code of Federal Regulations. The contractor will obtain appropriate AEC licenses, or agreement state licenses, as required by Title 10 CFR for receipt, manufacture, possession, use, storage, shipping, export and import as necessary. The contractor will also provide data necessary for the proper DA agency to obtain appropriate AEC licenses, or DA authorizations for Department of the Army receipt, possession, storage, use, distribution and maintenance. Intentionally added radioactive material is defined as radioactive material requiring specific

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licensing under the regulations issued pursuant to the Atomic Energy Act of 1954, as amended, as set forth in Title 10 CFR and any other radioactive material not requiring specific licensing in which the radioactivity per gram is greater than 0.002 microcurie.

3.16 Electrical characteristics.

3.16.1 Input power. The ADAS shall give full specified performance when operated from aircraft power sources with characteristics as defined in MIL-STD-704 having limits as specified below and tested in accordance with 4.8.1:

a. 20 to 29 Vdc, 7.0 amps max.

b. 105 to 125 Vac, 400 Hz ±20 Hz, 400 ma max.

3.16.2 Undervoltage protection. The equipment shall not be damaged by dc voltages below the minimum specified above and shall automatically resume normal operation when the voltage returns within limits. (See 4.8.2.)

3.17 Data annotation.

3.17.1 Fixed data input-output-annotation requirements. The thumbwheel indications on the CDM shall be identical to those displayed on the monitoring scope, CDM display CRT, and RHA when tested in accordance with 4.9 and 4.11. (IR filter data shall be simulated as an input coming from the AN/AAS-24( ). SLAR Range, SLAR Range Delay and SENSOR Identification shall also be simulated.) In addition, the total time test shall have an accuracy of ±2 seconds when tested in accordance with 4.9.5.2 and the Frame Number shall return to 0 when tested in accordance with 4.9.6. This requirement applies to all data listed below in either BCD, numeric or alternate mode except as indicated.

	Test Para.	Tolerance
Date (day, month, year)	4.9.1	±0
Sortie Number	4.9.2	±0
Taking Unit	4.9.4	±0

(NOTE: ±0 indicates repeat of input)

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	Test Para.	Tolerance
Time Insertion	4.9.5.1	±0
Time Display	4.9.5.2	±2 secs
Sensor Identification	4.9.11	±0
Frame Number Reset*	4.9.6	±0
Focal Length*	4.9.7	±0
KA-76 Angular Position*	4.9.8	±0
SLAR Range*	4.9.9	±0
SLAR Range Delay	4.9.10	±0
IR Filter	4.9.3	±0

(NOTE: ±0 indicates repeat of input)

*Numeric Only

3.17.2 Variable data accuracy. The AN/AYA-10( ) shall present the data listed in Table II to the cathode ray tube (CRT) of the CDM and the Recording Head Assemblies (RHA's) with the accuracies indicated below. The complete data block will be updated within 18 milliseconds. Internal timing shall preclude premature unblanking of the data block. (See 4.10 and 4.11.)

Function	BCD Format	Numeric Format	Ref Paragraph
Easting & Northings	±0.1 KM	±0.1 KM	4.10.10, 4.10.11
Heading	±0.1 Degree	±0.1 Degree	4.10.9
Roll	±0.3 Degree	±1.0 Degree	4.10.2
Pitch	±0.3 Degree	±1.0 Degree	4.10.3
Radar Altitude	±10 feet	±10 feet	4.10.7
Time	±1.0 second per 24 hour day	±1.0 second per 24 hour day	4.9.5.2
Barometric Altitude	±20.0 feet	Not applicable	4.10.1

NOTE: Tolerances shown indicate required conformance of output display to input signal.

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Function	BCD Format	Numeric Format	Ref Paragraph
Drift Angle	±0.1 Degree	Not applicable	4.10.8
Ground Speed	Not applicable	±1.0 Knot	4.10.4
Cycling Rate KA-60 (Forward and aft)	Not applicable	*±1/2 Frame/Sec	4.10.5.2, 4.10.6.2

*Readout to nearest integer only dependent upon voltages defined in 4.10.5.2 and 4.10.6.2.

3.18 Film annotation.

3.18.1 Data demand response. When tested in accordance with 4.12, the output display shall be in the mode selected as shown below and be presented only once for each demand pulse, except in the ALT mode, in which BCD and numerical display are presented alternately for each data demand. (See 4.12.)

Display Select	Data Demand	Mode
KA-60 (FWD)	KA-60 (FWD)	BCD
KA-60 (FWD)	KA-60 (FWD)	NUM
KA-60 (FWD)	KA-60 (FWD)	ALT
KA-60 (AFT)	KA-60 (AFT)	BCD
KA-60 (AFT)	KA-60 (AFT)	NUM
KA-60 (AFT)	KA-60 (AFT)	ALT
KA-76	KA-76	BCD
KA-76	KA-76	NUM
KA-76	KA-76	ALT
IR/SLAR	IR	BCD
	SLAR	BCD
IR/SLAR	IR	NUM
	SLAR	NUM
IR/SLAR	IR	ALT
	SLAR	ALT

3.18.2 Film exposure. Exposures will be taken with the following types of film using test procedures 4.13.1 thru 4.14.3 and shall be examined in accordance with 4.15 through 4.16.6. The measurements shall meet the requirements of MIL-STD-782.

DuPont 111R, or equal.
Eastman Kodak SO-201, or equal.

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PAN-X Eastman Kodak Commercial (slow), or equal.
PLUS-X PAN Eastman Kodak Commercial (medium), or equal.
TRI-X PAN Eastman Kodak Commercial (fast), or equal.

3.18.3 Skew compensation (KA-76).

3.18.3.1 Zero film-speed-compensation. The linear deviation as measured in accordance with 4.14.1 shall be less than ±0.024 inches.

3.18.3.2 Maximum film-speed-compensation. The reference dot displacement measured by comparing the exposed film for maximum film-speed-compensation (4.14.2) with the zero film-speed-compensation exposure (4.14.1) shall be 0.008 inch, +0.010 inch, -0.002 inch, toward the left.

3.18.3.3 Medium film-speed-compensation. The reference dot displacement measured by comparing the exposed film for medium film-speed-compensation (4.14.3) with the zero film-speed-compensation exposure (4.14.1) shall be 0.004 inch, +0.006 inch, -0.002 inch, toward the left.

3.18.4 General dot characteristics.

3.18.4.1 Dot shape and density. All dot shape and density characteristics when analyzed in accordance with the procedures of MIL-STD-782 shall meet the requirements of MIL-STD-782 including the acceptibility charts therein. (See 4.15.)

3.18.4.2 Minimum density. The minimum density difference between center dot and inter-dot density shall be in accordance with MIL-STD-782. (See 4.15.)

3.18.5 Data format.

3.18.5.1 Numeric annotation data block format. The numeric display shall consist of eight lines and eight characters per line. The characters shall not exceed .044 inches in height. Placement of the data block shall be consistent with MIL-STD-782 requirements for the BCD format. When the film is viewed emulsion down, the image of the numeric block will be right-handed. The numeric formats for the imaging sensors shall be in accordance with Figures 1 and 1a. (See 4.16.6.)

3.18.5.2 BCD annotation data block mensuration and format. The BCD mensuration and format shall be as described in MIL-STD-782 with the exceptions noted in Table II. (See 4.16 through 4.16.5.)

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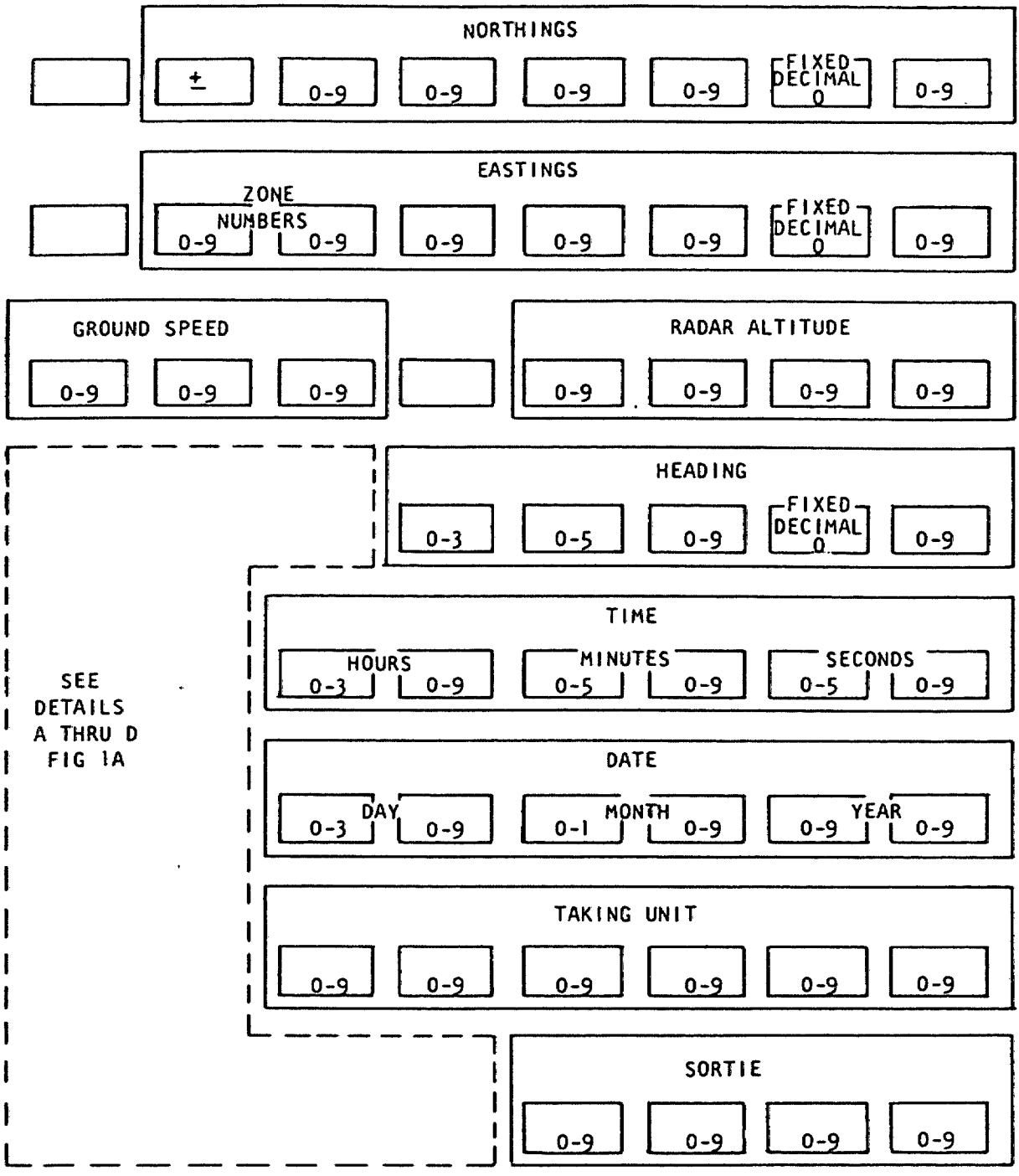
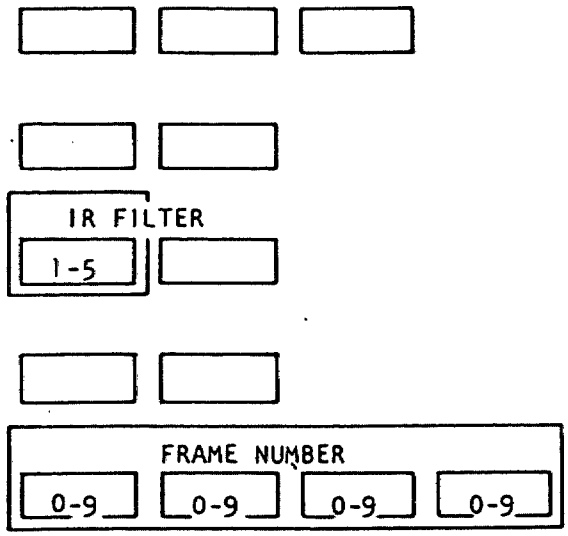
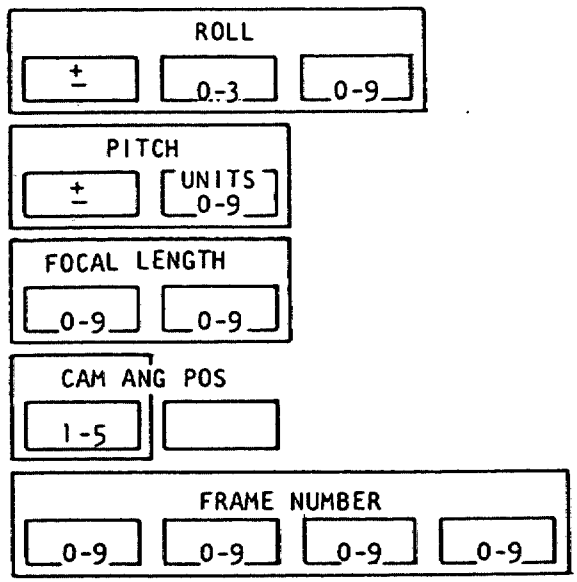


FIGURE 1
NUMERIC FORMAT

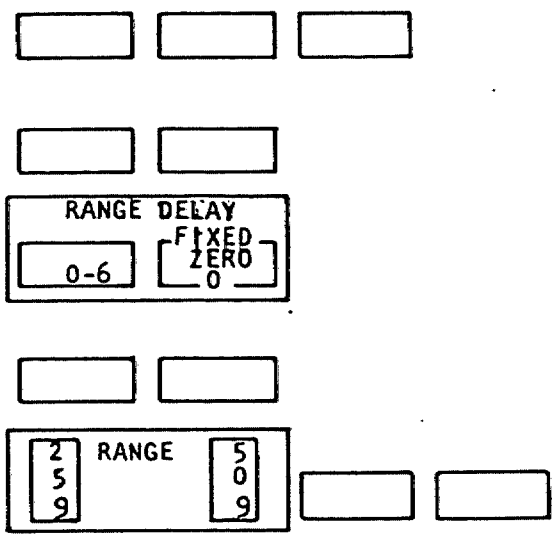
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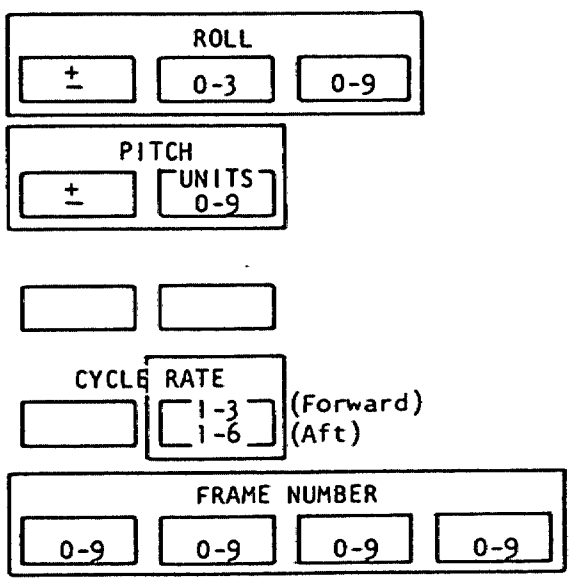
DETAIL A
IR DISPLAY



DETAIL B
KA 76 DISPLAY



DETAIL C
SLAR DISPLAY



DETAIL D
KA 60 DISPLAY

FIGURE 1A
NUMERIC FORMAT

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3.19 Environmental service conditions. The equipment shall meet the environmental service conditions specified below. The equipment shall have no deterioration or change in the tolerance limits specified in 3.17.1 and 3.17.2 or of any internal or external parts which could prevent it from meeting operational service or maintenance requirements after subjection to the following tests:

3.19.1 Acceleration. The equipment shall meet specified performance when exposed to steady-state acceleration levels of 6 g's when tested in accordance with 4.17.7. During testing, the equipment shall meet full specification performance for the following measurement:

Paragraph	Measurement
4.11	Functional Test

After testing, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
4.5.1	Group A inspection
4.5.2	Group B inspection

3.19.2 Shock (bench-handling). The equipment shall withstand shocks encountered in servicing as tested in accordance with 4.17.4a. After testing, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
4.5.1	Group A inspection
4.5.2	Group B inspection

3.19.3 Shock. During testing as specified in 4.17.4b the equipment shall meet full specification performance for the following measurement:

Paragraph	Measurement
4.11	Functional Test.

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After testing, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
4.5.1	Group A inspection
4.5.2	Group B inspection

3.19.4 Crash safety. There shall be no failure of mounting attachments and the dummy load shall remain in place and not create a hazard after the testing specified in 4.17.4c.

3.19.5 Vibration. The equipment shall not malfunction or suffer degradation when tested to the following requirements in accordance with 4.17.4d.

a. The equipment as normally mounted with vibration isolators in place, shall be subjected to the following amplitudes over the frequency range of 5 Hz to 500 Hz.

5 Hz	-	14 Hz	0.10 inches (D.A.)
14 Hz	-	23 Hz	1 g
23 Hz	-	74 Hz	0.036 inches (D.A.)
74 Hz	-	500 Hz	10 g

b. The equipment with vibration isolators removed shall be subjected to the following amplitudes over the frequency range of 5 Hz - 500 Hz.

5 Hz	-	20 Hz	0.10 inches (D.A.)
20 Hz	-	500 Hz	2 g

c. During testing the equipment shall meet full specification performance for the following measurement:

Paragraph	Measurement
4.11	Functional Test

After testing, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
4.5.1	Group A inspection
4.5.2	Group B inspection

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3.19.6 Temperature-altitude. During testing as specified in 4.17.2, the equipment shall meet full specification performance for the following measurement:

Paragraph	Measurement
4.11	Functional Test

After testing, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
4.5.1	Group A inspection
4.5.2	Group B inspection

3.19.7 Explosion. The equipment shall not cause an explosion at any of the test altitudes, when tested in accordance with 4.17.3. Equipment operation shall consist of the functional test of 4.11.

3.19.8 Fungus resistance. The equipment in the assembled condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.17.6. Only inherent fungus resistant grades of materials per Requirement 4, MIL-STD-454 shall be used. Prior to delivery, all fungus nutrient contaminants or residues shall be removed.

3.19.9 Humidity. During testing as specified in 4.17.5, the measurements to be taken during the 5 hour period specified therein shall consist of the following:

Paragraph	Measurement
4.11	Functional Test

After testing, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
4.5.1	Group A inspection
4.5.2	Group B inspection

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3.19.10 Electromagnetic compatibility. The AN/AYA-10( ) shall comply with the requirements of MIL-I-6181D and MIL-STD-826A as specified hereinafter. (See 4.18.)

3.19.10.1 Conducted interference, power leads MIL-I-6181D, paragraph 4.3.1.1. 28 VDC leads - As required by MIL-I-6181D. 115 V, 400 Hz - Narrowband - 2 dB deviation at 10 MHz, 4 dB deviation at 12 MHz on the high lead. 4 dB deviation at 10 MHz, 6 dB deviation at 12 MHz and 4 dB deviation at 14 MHz on the neutral lead. Broadband. As required by MIL-I-6181D.

3.19.10.2 Conducted interference, interconnecting leads MIL-I-6181D, paragraph 4.3.1.3 J2 - Broadband, 0.75 - 4.5 MHz with max deviation of 19 dB at 2.0 MHz. Broadband, 5.5 - 10 MHz with max deviation of 8 dB at 6.0 MHz. Broadband, 10.4 - 22 MHz with max deviation of 6 dB at 16 MHz. J3 - Broadband, 0.62 - 2.8 MHz with max deviation of 15 dB at 2.0 MHz. Broadband, 4.0 - 5.3 MHz with max deviation of 2 dB at 5.0 MHz. J4 - Broadband, 0.35 - 5.0 MHz with max deviation of 14 dB at 2.0 MHz. J5 - Broadband, 0.60 - 2.3 MHz with max deviation of 8 dB at 1.5 MHz. J6 - Broadband, 0.70 - 2.0 MHz with max deviation of 6 dB at 1.5 MHz. J7 - Broadband, 0.60 - 2.7 MHz with max deviation of 9 dB at 2.0 MHz. Broadband, 14 - 18 MHz with max deviation of 3 dB at 16 MHz. J9 - Broadband, 0.50 - 25 MHz with max deviation of 25 dB at 2.0 MHz. Narrowband, 16 dB deviation at 12 MHz.

3.19.10.3 Radiated interference, case and cables - MIL-I-6181D, paragraph 4.3.2. Broadband - 0.65 - 1.5 MHz with max deviations of 3 dB at 0.70 MHz, 4 dB at 1 MHz and 2 dB at 1.3 MHz. Broadband - 1.5 - 3 MHz with max deviation of 4 dB at 2 MHz. Broadband - 13 to 120 MHz with max deviations of 3 dB at 14 MHz, 4 dB at 20 MHz, 5 dB at 24 MHz, 29 dB at 30 MHz, 13 dB at 40 MHz, 24 dB at 58 MHz, 15 dB at 66 MHz and 8 dB at 90 MHz. Narrowband - 8 dB deviation at 12 MHz.

3.19.10.4 Radiated susceptibility, R.F. As required by MIL-I-6181D, paragraph 4.3.4.2.

3.19.10.5 R.F. conducted susceptibility, powerlines. As required by MIL-I-6181D, paragraph 4.3.4.1.1.

3.19.10.6 Audio conducted susceptibility, powerlines. As required by Method 5001, MIL-STD-826A - Frequency range of test shall be 30 Hz to 15 kHz.

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3.19.10.7 Spike susceptibility, powerlines, Method 5006.1. As required by MIL-STD-826A.

3.19.10.8 Radiated susceptibility, induction field. As required by Method - 6002, MIL-STD-826A. (See 4.18.1.)

3.20 Bonds, grounds and connectors. Bonding for equipment current path return, r.f. potentials and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case mounted connectors shall have an approved r.f. conductive finish and shall be installed as above. Bonding impedance of the connectors shell to the case shall not exceed 2.5 milliohms. If interconnecting cables require over-all shields, the installation document prepared by the contractor shall specify such cables and indicate that shielded cables shall be terminated with cable connectors having an approved r.f. conductive finish and provided with EMI back shells for peripheral bonding of the cable shield. Specification MIL-B-5087 shall be used to the fullest extent in determining application of bonds, grounds and case mounted connectors. (See 4.19.)

3.21 Interchangeability. Like units, assemblies, sub-assemblies and replaceable parts shall conform to MIL-STD-454, Requirement 7. (See 4.21.)

3.22 Workmanship. The equipment shall be manufactured and assembled in accordance with paragraph 3.5, MIL-E-5400. (See 4.25.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (Does not include preparation for delivery) (See 4.3).

b. Inspections covered by subsidiary documents. (See 4.4.)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)

(2) Quality conformance inspection of preparation for delivery. (See 4.28.)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish 3 first article units of the complete AN/AYA-10( ).

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning 4.6, burn-in 4.22 and Groups A, B and C inspections (4.5.1 through 4.5.4) in that order. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A and B inspections.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirements specified in the contract or purchase order:

Item	Where required
Parts, materials and processes; general	3.4
Finish	3.6
Marking	3.7 - 3.7.3

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4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in the subsidiary documents covering the items listed in 4.4, preconditioning 4.6, burn-in 4.22 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. All components and subassemblies of the AN/AYA-10( ) system, which may have been replaced as a result of failures during any type of testing (i.e., Groups A, B and C, reliability, etc.) shall have met the minimum burn-in requirements of 4.22b and c respectively before preparation for delivery. Burn-in of such replaced components and sub-assemblies may be accomplished separately from the equipment in which delivered. In addition, such burn-in of replacement components and sub-assemblies need not be accomplished as part of any AN/AYA-10( ) system but may be done on an individual component or sub-assembly basis.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table III.

Table III. Group A Inspection

Inspection	Requirement Paragraph	Test Paragraph	AQL Major	AQL Minor
Visual and Mechanical				
CDM C-8338( )/AYA-10	3.22	4.25	1.0%	4.0%
SDC CV-2647( )/AYA-10	3.22	4.25	1.0%	4.0%
Mount, MT-4539( )/AYA-10	3.22	4.25	1.0%	4.0%
Safety	3.15	4.24.1	1.0%	4.0%
Electrical Tests				
Input power	3.16.1	4.8.1	1.0%	
Under voltage protection	3.16.2	4.8.2	1.0%	*
Bonds and grounds	3.20	4.19	1.0%	

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Inspection	Requirement Paragraph	Test Paragraph	AQL Major	AQL Minor
Data Annotation				
Fixed Data				
Date	3.17.1	4.9.1, 4.11		
Sortie number	3.17.1	4.9.2, 4.11		
IR filter	3.17.1	4.9.3, 4.11		
Taking unit	3.17.1	4.9.4, 4.11		
Time insertion	3.17.1	4.9.5.1, 4.11	1.0% for the group	*
Time display	3.17.1	4.9.5.2, 4.11		
Frame number reset	3.17.1	4.9.6, 4.11		
Focal length	3.17.1	4.9.7, 4.11		
KA-76 angular position	3.17.1	4.9.8, 4.11		
SLAR range	3.17.1	4.9.9, 4.11		
SLAR range delay	3.17.1	4.9.10, 4.11		
Sensor identification	3.17.1	4.9.11, 4.11		
Variable Data				
Barometric altitude	3.17.2	4.10.1, 4.11		
Roll	3.17.2	4.10.2, 4.11		
Pitch	3.17.2	4.10.3, 4.11	1.0% for the group	*
Ground speed	3.17.2	4.10.4, 4.11		
Cycling rate	3.17.2	4.10.5, 4.10.6, 4.11		
Radar altitude	3.17.2	4.10.7, 4.11		
Drift angle	3.17.2	4.10.8, 4.11		
Heading	3.17.2	4.10.9, 4.11		
Northing	3.17.2	4.10.10, 4.11		
Easting	3.17.2	4.10.11, 4.11		

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Table III. Group A Inspection (Cont.)

Inspection	Requirement Paragraph	Test Paragraph	AQL Major	AQL Minor
Data Demand Response	3.18.1	4.12	1%	*
CDM CRT Display Intensity	3.11	4.26	1%	*
Vg/H Output	3.9.2	4.10.5, 4.10.6	1%	*
Command Navigation Data Request Output	3.9.3	4.10.12	1%	*
Self Test	3.10	4.27	1%	*

*All electrical and operational defects are considered major

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans for the AQL's listed in Table IV shall be as follows:

AQL	Inspection Level
4.0 percent	S-4
6.5 percent	S-4

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

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Table IV. Group B Inspection

Inspection	Requirement Paragraph	Test Paragraph	AQL
Film Exposure Tests			
IR tests	3.18.2	4.13.1	4.0% for the group
SLAR tests	3.18.2	4.13.2	
KA-76 slow film test	3.18.2	4.13.3	
KA-76 medium film test	3.18.2	4.13.4	
KA-76 fast film test	3.18.2	4.13.5	
KA-60 slow, medium and fast film test	3.18.2	4.13.6	
KA-76 Skew Compensation			
Zero film speed compensation	3.18.3.1	4.14.1	
Maximum film speed compensation	3.18.3.2	4.14.2	
Medium film speed compensation	3.18.3.3	4.14.3	
General Dot Characteristics	3.18.4	4.15	
Data Format Measurement	3.18.5	4.16 thru 4.16.6	
Interchangeability			
Mechanical	3.21	4.21	6.5%

4.5.3.1 Sampling for Group C inspection. One sample unit shall be selected at random for each group of inspections in Table V from each 50 or fraction thereof produced. Thus, Group C testing shall be performed on samples selected from any fraction of 50 units that remains beyond the first 50 units or on samples from all succeeding multiples of 50 units or fraction thereof. Initial samples for the first Group C production tests will be selected at random from a production lot near the end of the first 50 produced. Samples for remaining Group C production tests will be selected at random from production lots near the end of each subsequent 50 units produced or any fraction thereof.

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4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in accordance with Table V.

Table V. Group C Inspection

Inspection	Requirement Paragraph	Test Paragraph
Group 1		
Temperature - Altitude	3.19.6	4.17.2
Humidity	3.19.9	4.17.5
Group 2		
Explosion	3.19.7	4.17.3
Acceleration	3.19.1	4.17.7
Vibration	3.19.5	4.17.4d
Shock	3.19.3	4.17.4b
Shock (bench-handling)	3.19.2	4.17.4a
Crash safety	3.19.4	4.17.4c
Fungus*	3.19.8	4.17.6
Group 3		
EMI/EMC	3.19.10	4.18

NOTES: (1) The tests above the broken lines are considered non-damaging. The order of these tests and choice of units to use may be varied if approved. However, the same sequence of testing shall be used for first article and production Group C testing.

(2)* Following fungus testing, the test item shall be thoroughly washed, cleaned, dried, and refurbished if necessary before proceeding with subsequent tests.

(3) For any particular Group C test unit, satisfactory completion of Group A and B tests, where specified at the completion of environmental tests, shall satisfy the requirement for Groups A and B testing prior to commencement of the next environmental test. The exception is where the Government representative has determined that equipment handling or other conditions between tests can be expected to invalidate the prior Groups A and B test results.

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4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the parts and materials, as applicable, and on all units of product which are manufactured under the same conditions and which the Government considers subject to the same failures. Reports shall be forwarded through the Government quality assurance representative to the responsible technical activity designated in the contract. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the technical activity). However, final acceptance and shipment will be withheld until Group C reinspection results have shown that the corrective action was effective and was approved by the technical activity. Additional actions required relative to Group C failures shall be as specified in the contract.

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and meet Group A and B inspection.

4.6 Vibration preconditioning. All AN/AYA-10( ) systems shall be vibrated (without vibration isolators) for a total period of 30 minutes prior to being subjected to the burn-in of 4.22. The equipment shall be operated continually during this period only to detect flaws and poor workmanship. Operation within specified limits of performance shall not be required. The direction of vibration shall be in three planes; vertical to the normal mounting plane for 10 minutes, lateral to the plane for 10 minutes, and fore and aft for 10 minutes.

Vibration	33 - 100 cps at ±2.2 g's and .33 oct/min
Temperature	Standard environmental (see 4.7.1)
Humidity	Standard environmental (see 4.7.1)

4.7 Tests, general.

4.7.1 Standard environmental conditions. The following conditions shall be used to establish satisfactory performance characteristics under standard environmental conditions when the latter are specified.

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a.	Temperature	73° ± 18°F
b.	Altitude	28.5, +2.0, -4.5 inches of mercury
c.	Vibration	None
d.	Relative humidity	50% ± 30%
e.	Input power	28 ±.5 Vdc, 7.0 amps maximum 115 ±1 Vac, 400 ±10 Hz, 400 ma maximum.

4.7.2 Test set-up and equipment.

4.7.2.1 Bench test set-up. A bench test set-up which includes test equipment equivalent to that listed in 4.7.2.2 and 4.7.2.3 shall be used for monitoring signal inputs and outputs of the AN/AYA-10( ). Simulated digital (+3 BCD), analog and synchro signals shall be provided to perform the tests specified herein (including + and - sign signals when necessary) and to assure that the AN/AYA-10( ) will interface properly within the operating limits of the associated equipment in 3.1.1. The operating limits of the input data from the associated equipment are given in Table I. In addition simulated predetermined special data signals shall be provided in conjunction with the CDM output signals to perform the functional test specified in 4.11. A data list of voltages, waveforms and tolerances that apply during testing shall be furnished by the contractor prior to testing as part of the first article test plan (4.3.3) as assurance that the interfacing referred to above is correct.

4.7.2.2 Commercial test equipment. (Contractor furnished).

a. Textronix 503 Oscilloscope or equal.

b. Textronix 545 Oscilloscope or equal.

c. Fluke 803 Differential Voltmeter or equal.

d. Sorensen FCD 500 Power Supply (115 V 400 Hz) or equal.

e. Harrison Lab 510A Power Supply (+28 Vdc) or equal.

f. Praktina 35 mm camera with f/2 lens or Modified Speed Graphic 4" x 5" camera with f/2 lens, or equal.

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g. AnscoModel 4 Microdensitometer or equal.

h. Zeiss 20-50-00 X-Y Microscope or equal.

4.7.2.3 Special test equipment. (Contractor furnished).

a. Recording Head Assemblies (RHA's) for displaying AN/AYA-10( ) numeric and BCD outputs.

b. Adapter, RHA to camera.

4.8 Electrical tests.

4.8.1 Input power check. Turn the CDM power switch to ON. Check and readjust, if necessary, the +28 Vdc and 115 Vac input voltages. With the simulated data input of 4.11, observe and check presence of correct numeric and BCD displays on the CDM monitor. Decrease the dc and ac voltages to 20 Vdc and 105 Vac respectively. There shall be no change in the displays and the cooling fan shall function normally. Increase the dc and ac voltages to 29 Vdc and 125 Vac respectively. There shall be no change in the displays and the cooling fan shall function normally. (See 3.16.1.)

4.8.2 Undervoltage test. Reduce the dc voltage to 10 Vdc for 30 seconds. The voltage shall then be raised to 28 Vdc. The presence of the numeric and BCD displays shall be observed and shall be the same as 4.8.1. (See 3.16.2.)

4.9 Fixed data tests. Fixed data is obtained by setting the thumbwheel switches on the Control Display Monitor (CDM) and observing the data of the digital displays. For each reading check the displays of the monitoring equipment, (oscilloscope, RHA, and CDM) for identical presentation.
NOTE: Simulated data demand signals are required for presentation.

4.9.1 Date. On the CDM, set DAY, MONTH, and YEAR thumbwheel switches to position 11-11-11 through 39-19-99 by increasing both digits in the year column by 1, the least significant digit (LSD) of the month and day by 1, and the most significant digit of the day by 1 up to 3 until 39-19-99 is reached and then include 00-00-00. Observe appropriate changes on the monitoring equipments by flipping MODE switch to NUM then to BCD. (See 3.17.1.)

4.9.2 Sortie No. On the CDM, set SORTIE thumbwheel switches to positions 0000 through 9999 by increasing all digits by 1 for each change until

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9999 is obtained. Observe appropriate changes on the monitoring equipments by flipping MODE switch to NUM then BCD. (See 3.17.1.)

4.9.3 IR Filter. Simulate the five IR FILTER inputs and observe appropriate changes on the monitoring equipments by flipping the MODE switch to NUM then to BCD. (See 3.17.1.)

4.9.4 Taking Unit. On the CDM, set TAKING UNIT thumbwheel switches to positions 000000 to 999999 by increasing all digits by 1 until 999999 is obtained. Observe appropriate changes on the monitoring equipments by flipping MODE switch to NUM then to BCD. (See 3.17.1.)

4.9.5 Time.

4.9.5.1 Time Insertion. On the CDM, set HRS, MINS, and SECS thumbwheel switches to positions 000000 through 39-59-59 by increasing the LSD of each pair by 1 and the MSD of the seconds and minutes by 1 through 5 and the hours MSD through 3 until 39-59-59 is obtained. Observe appropriate changes on the monitoring equipments by flipping the MODE switch to NUM and pushing TIME SET switch (if true data up-dates before complete reading is taken, re-insert by pressing TIME SET switch). Flip CDM switch to BCD, push CDM TIME SET switch. Press PUSH TO TEST switch and hold to obtain readings on CDM. (See 3.17.1.)

4.9.5.2 Time Display. On the CDM, set HRS, MINS, and SECS thumbwheel switches to the approximate correct local time. Insert the time by pressing TIME SET switch, and simultaneously observing a clock: Observe on the monitoring equipments and check that the readings are correct and continuously updating. Total time display test requires two hours. However, during this time remaining portion of test may be continued. (Caution - Removing system power invalidates the time display test. Use care in operating the remaining tests so not to accidentally re-insert TIME SET until this portion of the test is complete). After two hours of continuous time of test observe on the monitoring equipment the reading of time to the nearest second. Simultaneously compare these readings with the starting clock readings and note the difference between the two readings. The difference shall meet the requirements of 3.17.1.

4.9.6 Frame Number Reset. Observe Frame Number of numeric displays by flipping CDM MODE switch to NUM position. While observing the monitoring equipment push FRAME NO. RESET switch. Observe and check that the frame number data on the display return to zero. (See 3.17.1.)

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4.9.7 Focal Length. On the CDM set KA-76 EXP to 5 and set FOCAL LENGTH thumbwheel switches and observe appropriate change on the monitoring equipments by flipping MODE switch to NUM. (See 3.17.1.)

4.9.8 KA-76 Angular Position. Simulating KA-76 angular positions 1, 2, 3, 4, and 5, observe appropriate changes on the monitoring equipment flipping the MODE switch on the CDM to NUM. (See 3.17.1.)

4.9.9 SLAR Range. Simulating SLAR Ranges of 25, 50 and 99 kilometers, observe appropriate changes on the monitoring equipments by flipping the MODE switch on the CDM to NUM. (See 3.17.1.)

4.9.10 SLAR Range Delay. Simulate SLAR Range Delays of 00, 10, 20, 30, 40, 50, and 60 kilometers. Observe appropriate changes on the monitoring equipments by flipping the MODE switch on the CDM to NUM, then to BCD. (See 3.17.1.)

4.9.11 SENSOR Identification. Simulate SENSOR Identifications by varying applicable simulated inputs to obtain the outputs defined in Table II. Observe changes on the monitoring equipments by flipping MODE switch to BCD. Display accuracies shall conform to the tolerances of 3.17.2.

4.10 Variable data tests. Variable data is obtained by applying simulated inputs and observing the data on the displays. For each reading, check the displays of the monitoring equipment (oscilloscope, RHA and CDM). (See 3.17.2.)

4.10.1 Barometric Altitude. On the CDM, set the MODE switch to BCD. Simulate Barometric Altitudes of 0 ft, 500 ft, 1000 ft, 2000 ft, 4000 ft, 6000 ft, 8000 ft and 9950 ft. Observe the Barometric Altitudes on the monitoring equipment. (See 3.17.2.)

4.10.2 Roll. Simulate a Roll of -30° and progress in 5° steps up thru 0° to $+30^{\circ}$ (13 readings). Observe the roll data on the monitoring equipment by flipping CDM MODE switch to BCD then to NUM. (See 3.17.2.)

4.10.3 Pitch. Simulate a Pitch of -30° and progress in 5° steps up thru 0° to $+30^{\circ}$ (13 readings). Observe the Pitch data on the monitoring equipment by flipping CDM MODE switch to BCD. For numeric display simulate from -9° thru 0 to $+9^{\circ}$ and observe output in NUM mode. (See 3.17.2.)

4.10.4 Ground Speed. Simulate Ground Speeds of 000 to 999 knots (10 readings) by increasing all digits by 1. Observe appropriate changes on the monitoring equipment by flipping the CDM MODE switch to NUM. (See 3.17.2.)

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4.10.5 Vg/H Output (KA-60( ) FWD, KA-76( ) and AN/AAS-24( ) IR) and Cycling Rate (KA-60 FWD).

4.10.5.1 Vg/H DC Analog Output Voltage (KA-60( ) FWD, KA-76( ) and AN/AAS-24( ) IR. Simulate inputs corresponding to the Vg/H ratios shown below. Measure Vg/H output voltages as indicated below and determine that their linearity conforms to 3.9.2.1.

Vg/H Ratio	Vg/H Output Voltage (Vdc ± 5%)
2.00	100.6
1.75	88.0
1.35	67.9
0.70	35.2
0.20	10.1
0.01	0.5

4.10.5.2 Cycling Rate (KA-60 FWD). Provide input voltages throughout the ranges shown below and observe the cycling rate changes on the monitoring equipment by setting the CDM MODE switch to NUM. (See 3.9.2.2.)

Input Voltage (Vdc)	Cycling Rate Display
0.0 to 7.0	BLANK
24.1 to 38.1	1
55.3 to 69.3	2
86.4 to 100.4	3

4.10.6 Vg/H Output (KA-60 AFT) and Cycling Rate (KA-60 AFT).

4.10.6.1 Vg/H DC Analog Output Voltage (KA-60 AFT). Simulate inputs corresponding to the Vg/H ratios shown below. Measure Vg/H output voltages as indicated below and determine that their linearity conforms to 3.9.2.1.

Vg/H Ratio	Vg/H Output Voltage (Vdc ± 5%)
2.00	50.3
1.00	50.3
0.84	42.3
0.68	34.2
0.50	25.2
0.34	17.1
0.18	9.1
0.01	0.5

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4.10.6.2 Cycling Rate (KA-60 AFT). Provide input voltages throughout the ranges shown below and observe the cycling rate changes on the monitoring equipment by setting the CDM MODE switch to NUM. (See 3.9.2.2.)

Input Voltage (Vdc)	Cycling Rate Display
0.0 to 2.0	BLANK
6.9 to 10.9	1
15.9 to 19.9	2
24.8 to 28.8	3
33.7 to 37.7	4
42.7 to 46.7	5
51.6 to 55.6	6

4.10.7 Radar Altitude. Simulate Radar Altitude from 0000 thru 9999 feet by changing all digits by 1 (9999 to 8888, etc.). Observe appropriate changes on the monitoring equipment by flipping the CDM MODE switch to NUM, then to BCD. (See 3.17.2.)

4.10.8 Drift Angle. Simulate Drift Angles of 0000 thru 9999 degrees (10 readings) by changing all digits by 1. Set the CDM MODE switch to BCD. Observe appropriate changes on the monitoring equipment. (See 3.17.2.)

4.10.9 Heading. Simulate Headings of 0000 to 9999 degrees (10 readings) by increasing all digits by 1. Observe appropriate changes on the monitoring equipment by flipping the CDM MODE switch to NUM, then to BCD. (See 3.17.2.)

4.10.10 Northing. Simulate Northings of 0000.0 to 9999.9 kilometers (10 readings) by increasing all digits by 1. Observe appropriate changes on the monitoring equipment by flipping the CDM MODE switch to NUM, then to BCD. (See 3.17.2.)

4.10.11 Easting. Simulate Eastings (UTM and Zone Number) from 000.0 to 999.9 kilometers (10 readings) and 00 to 99 (10 readings) respectively by increasing all digits by 1. Observe appropriate changes on the monitoring equipment by flipping the CDM MODE switch to NUM, then to BCD. (See 3.17.2.)

4.10.12 Command navigation data request output. During one of the above simulations, measure the data request signal to the INS at the 3 line output to the INS. Each line output shall consist of four 32 microsecond pulses in each 18.2 milliseconds at a nominal voltage of 5 volts $\pm$0.5 volts in the coding which corresponds to the selected parameter, i.e., northing, heading, etc. (See 3.9.3.)

4.11 Functional test. Simulate data input, including signs and decimals, in conjunction with CDM output signals to test all data positions in BCD and

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numeric format as shown in Table II. The outputs displayed on the monitoring equipment shall be the correct predetermined values and shall meet the tolerances of 3.17.1 and 3.17.2. In addition, no degradation in presentation shall be evident. (See 3.17.1, 3.17.2.)

4.12 Data demand response. Simulate a data demand single pulse and an INS data ready signal. Set the CDM MODE switch to BCD and select display according to chart below. Pulse ten times at random intervals, observe that the monitoring equipment responds exactly i.e. only one display in response to each data demand. The display shall be a BCD display. Set the MODE switch to NUM and repeat pulse ten times. The display will be numeric. Set the MODE switch to ALT and repeat pulse ten times. The display will alternate between numeric and BCD. (See 3.18.1.)

DISPLAY SELECT	DATA DEMAND	MODE
KA-60 (FWD)	KA-60 (FWD)	BCD
KA-60 (FWD)	KA-60 (FWD)	NUM
KA-60 (FWD)	KA-60 (FWD)	ALT
KA-60 (AFT)	KA-60 (AFT)	BCD
KA-60 (AFT)	KA-60 (AFT)	NUM
KA-60 (AFT)	KA-60 (AFT)	ALT
KA-76	KA-76	BCD
KA-76	KA-76	NUM
KA-76	KA-76	ALT
IR/SLAR	IR	BCD
	SLAR	BCD
IR/SLAR	IR	NUM
	SLAR	NUM
IR/SLAR	IR	ALT
	SLAR	ALT

4.13 Film exposure tests. A total of 22 exposures will be made of the displays on recording head assemblies as indicated below using both BCD and numeric displays. Film measurements, where applicable will be in accordance with MIL-STD-782. Exposures will be taken with the following types of film:

Annotation of IR data on Dupont 111 R or equal.
Annotation of SLAR data on Eastman Kodak SO-201 or equal.
Annotation of KA-76, KA-60 (FWD) and KA-60 (AFT) data on:

slow:	PAN X Eastman Kodak Commercial or equal
medium:	PLUS-X PAN Eastman Kodak Commercial or equal
fast:	TRI-X PAN Eastman Kodak Commercial or equal

Turn OFF the DC switch to the CDM.

Install the RHA into the camera adapter.

Turn on the DC switch to the CDM.

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4.13.1 IR tests. Set the CDM Mode to BCD. Install Dupont 111 R film in the camera. Open the shutter, while simulating a single pulse data demand with an IR display. Close the camera shutter. Advance the film and set the CDM mode to NUM. Open the shutter and simulate a single pulse data demand with an IR display. Close the camera shutter. After the films have been developed, they shall be analyzed as specified in 4.15 through 4.16.6. (See 3.18.2.)

4.13.2 SLAR test. Set the CDM Mode to BCD. Install Eastman Kodak SO-201 film in the camera. Open the shutter, while simulating a single pulse data demand with a SLAR display. Close the camera shutter. Advance the film and set the CDM Mode to NUM. Open the shutter and simulate a single pulse data demand with a SLAR display. Close the camera shutter. After the films have been developed, they shall be analyzed as specified in 4.15 through 4.16.6. (See 3.18.2.)

4.13.3 KA-76 slow film test. Set the CDM Mode to BCD and set the KA-76 EXP thumbwheel on 7. Install PAN-X film in the camera. Open the shutter, while simulating a single pulse data demand, with a KA-76 display. Close the camera shutter. Advance the film and set the CDM Mode to NUM. Open the shutter and simulate a single pulse data demand, with a KA-76 display. Close the camera shutter. After the films have been developed, they shall be analyzed as specified in 4.14 through 4.16.6. (See 3.18.2.)

4.13.4 KA-76 medium film test. Set the CDM Mode to BCD and set the KA-76 EXP thumbwheel on 4. Install PLUS-X PAN film in the camera. Open the shutter, while simulating a single pulse data demand, with a KA-76 display. Close the camera shutter. Advance the film and set the CDM Mode to NUM. Open the shutter and simulate a single pulse data demand, with a KA-76 display. Close the camera shutter. After the films have been developed, they shall be analyzed as specified in 4.15 through 4.16.6. (See 3.18.2.)

4.13.5 KA-76 fast film test. Set the CDM Mode to BCD and set the KA-76 EXP thumbwheel on 3. Install TRI-X PAN film in the camera. Open the shutter, while simulating a single pulse data demand, with a KA-76 display. Close the camera shutter. Advance the film and set the CDM Mode to NUM. Open the shutter and simulate a single pulse data demand with a KA-76 display. Close the camera shutter. After the films have been developed, they shall be analyzed as specified in 4.15 through 4.16.6. (See 3.18.2.)

4.13.6 KA-60 slow, medium and fast film tests. Repeat 4.13.3, 4.13.4 and 4.13.5 except the KA-60-1 thumbwheels shall be set in 7, 4 and 3 respectively, in place of KA-76, and a single pulse data demand with a KA-60-1 display shall apply for each paragraph. Repeat for KA-60-2. (See 3.18.2.)

4.14 KA-76 skew compensation.

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4.14.1 Zero film speed compensation. Using the X-Y microscope and the film developed from paragraph 4.13.3, measure the linear deviation of the center of the index dot of column Z, line 32 of the Code Matrix block from a line through the center of the index dot of column Z, line 1, and perpendicular to a best fit straight line through line 32 of column X, Y and Z. This should be less than ±0.024 inches. (See 3.18.3.1.)

4.14.2 Maximum film speed compensation. Repeat paragraph 4.13.3 (BCD mode only) except simulate a ground speed of 200 knots. Measure the linear deviation on the new exposure as in paragraph 4.14.1, and compare with the exposure measured in that paragraph (zero compensation). The dot should be displaced toward the left by 0.008 inch (+.010 inch, and -.002 inches). (See 3.18.3.2.)

4.14.3 Medium film speed compensation. Repeat paragraph 4.13.3 (BCD mode only) except simulate a ground speed at 100 knots. Measure the linear deviation on the new exposure as in paragraph 4.14.1, and compare with the exposure measured in that paragraph (zero compensation). The dot should be displaced toward the left by 0.004 inch (+.006, -.002 inch). (See 3.18.3.3.)

4.15 General dot characteristics. On the BCD films, measure the size of four dots with a 2-mil (0.002 inch) aperture microdensitometer. With a recording microdensitometer, measure the density difference between the dot peaks and the interdot density. Scan the isolated dot with the 2-mil recording microdensitometer. The envelope of acceptability shown in MIL-STD-782 shall be superimposed on the above recording. The recording shall fall within this envelope except that a single spike no more than 0.0015 inches in diameter may extend into area A. The isolated dot microdensitometer recording shall be measured to find peak dot density to background density difference. (See 3.18.4.)

4.16 Data format measurement. With optical comparator or X-Y Microscope, make the following BCD film measurements on all films developed from paragraphs 4.13.1 through 4.13.6 using procedures outlined in MIL-STD-782. (See 3.18.5.2.)

4.16.1 Overall BCD code block dimensions shall be in accordance with MIL-STD-782 as measured with the X-Y microscope. (See 3.18.5.2.)

4.16.2 Reference line A is a best fit straight line that equalizes the positive and negative excursions of the bits. No excursion may be more than ±0.002 inches. Measure maximum excursion in 10^{-3} inches for each exposure listed below. (See 3.18.5.2.)

Exposure

IR	KA-76 Fast	KA-60 (AFT) Slow
SLAR	KA-60 (FWD) Slow	KA-60 (AFT) Med Speed
KA-76 Slow	KA-60 (FWD) Med Speed	KA-60 (AFT) Fast
KA-76 Med Speed	KA-60 (FWD) Fast	

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4.16.3 Within given digits: Column X, line 24;column Y, line 21; and column Z, line 13; no bit dot center shall depart more than ±0.002 inches from a line drawn through the center of its index dot parallel to reference line A. Measure maximum excursions in 10^{-3} inches for each exposure listed below. (See 3.18.5.2.)

Exposure
IR
SLAR
KA-76 Slow
KA-76 Med Speed
KA-76 Fast
KA-60 (FWD) Slow
KA-60 (FWD) Med Speed
KA-60 (FWD) Fast
KA-60 (AFT) Slow
KA-60 (AFT) Med Speed
KA-60 (AFT) Fast

4.16.4 No index dot center in column X, Y and Z shall depart more than ±0.012 inches from a line parallel to reference line B through the index dot files of column X or column Z. Measure maximum excursions in 10^{-3} inches for each exposure. (See 3.18.5.2.)

Exposure
IR
SLAR
KA-76 Slow
KA-76 Med Speed
KA-76 Fast
KA-60 (FWD) Slow
KA-60 (FWD) Med Speed
KA-60 (FWD) Fast
KA-60 (AFT) Slow
KA-60 (AFT) Med Speed
KA-60 (AFT) Fast

4.16.5 No index dot center shall be more than 0.003 inches greater distance from reference line B than an adjacent dot center. Measure maximum excursions in 10^{-3} inches for each case listed below. (See 3.18.5.2.)

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Exposure

IR
SLAR
KA-76 Slow
KA-76 Med Speed
KA-76 Fast
KA-60 (FWD) Slow
KA-60 (FWD) Med Speed
KA-60 (FWD) Fast
KA-60 (AFT) Slow
KA-60 (AFT) Med Speed
KA-60 (AFT) Fast

4.16.6 With an optical comparator or X-Y microscope, make the following numeric data block film measurements: Measure the size of two characters; column 4, line 4, and column 1, line 1 taking the top right corner of the display as column 1, line 1. No character shall exceed 0.044 inches in height. Measure for each exposure listed below. (See 3.18.5.1.)

Exposure

IR
SLAR
KA-76 Fast
KA-76 Med Speed
KA-76 Slow
KA-60 (FWD) Fast
KA-60 (FWD) Med Speed
KA-60 (FWD) Slow
KA-60 (AFT) Fast
KA-60 (AFT) Med Speed
KA-60 (AFT) Slow

4.17 Environmental tests. All test methods and procedures indicated below refer to MIL-STD-810.

4.17.1 Tolerances. Unless otherwise specified, the following tolerances shall be maintained:

Vibration amplitude	±10%
Vibration frequency	±2%
Shock amplitude	±15%

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Shock duration	±15%
Temperature	±2°C
Humidity	+5%, -0%
Altitude	±5%

4.17.2 Temperature-altitude. The equipment shall be subjected to the test of Method 504, Procedure I for Class 1A equipment in accordance with Table 504-II. The temperatures for Steps 5 and 7 shall be +81°C and +71°C respectively. The lowest and highest input voltages shall be the minimum and maximum voltage limits respectively specified in 3.16.1. (See 3.19.6.)

4.17.3 Explosion. The equipment shall be subjected to the test of Method 511, Procedure I. (See 3.19.7.)

4.17.4 Shock and vibration.

a. The equipment shall be subjected to the bench-handling shock test of Method 516.1, Procedure V. (See 3.19.2.)

b. The equipment shall be subjected to the shock test of Method 516.1, Procedure I. The shock pulses shall be in accordance with Figure 516.1-2 for flight vehicle equipment. (See 3.19.3.)

c. A dummy load representative of the equipment shall be subjected to the crash safety test of Method 516.1, Procedure III. The shock pulses shall be in accordance with Figure 516.1-2 for flight vehicle equipment. (See 3.19.4.)

d. The equipment shall be subjected to the vibration test of Method 514.1 and shall consist of Procedure I, Figure 514.1-2, Curve D (Part 1) and also Curve B (Part 2). The time schedules shall be in accordance with Table 514.1-II. (See 3.19.5.)

4.17.5 Humidity. The equipment shall be subjected to the test of Method 507, Procedure II. In step 6, measurements shall be taken during the last 5 hours of the last cycle. (See 3.19.9.)

4.17.6 Fungus test. The equipment, in the assembled condition, shall be subjected to 28 days of Method 508, MIL-STD-810. The equipment test model shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. If it is necessary for the manufacturer after assembly to remove accumulated production and handling contaminants by cleaning prior to packaging and delivery, the equipment shall

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not be tested for three days after cleaning to allow for complete evaporation of the cleaning compound. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 14 and 28 days of test, the control items per para. 3.1.3 and 3.1.4 of Method 508, MIL-STD-810 shall show profuse growth over at least 50% of the area of the control items. After 28 days, based upon visual examination, the equipment assembly shall pass if it shows no more than only sparse microbial growth with restricted tuberal growth development in an area of 1 to 10% or less of the total area and no more than six un-related minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacings, printed circuit boards, etc; with sparse growth due to random contamination or traces of unmixed material ingredients. No operational test is required after the fungus test to pass the test. The equipment shall fail if it shows more than the growth specified above. The test shall be repeated if the control items fail to show profuse growth after 14 and 28 days of test. (See 3.19.8.)

4.17.7 Acceleration. The equipment shall be subjected to the test of Method 513.1, Procedure II. The test level shall be 6 g's. (See 3.19.1.)

4.18 Electromagnetic compatibility. The AN-AYA-10( ) shall be tested for compliance to the requirements of 3.19.10. Test procedures shall be in accordance with MIL-I-6181D and MIL-STD-826A and the contractor prepared government approved test plan. (See 3.19.10.)

4.18.1 Radiated susceptibility induction field. The AN/AYA-10( ) system shall be tested in accordance with Method 6002, MIL-STD-826A. Twenty ampere, 400 Hz current shall flow in the wires wrapped around each case of the test sample and each interconnecting, input and output cable. (See 3.19.10.8.)

4.19 Bonding impedance. Prior to the performance of the EMC tests of 4.18, bonding impedances shall be measured for compliance to the requirements of 3.20. All measurement readings shall be recorded and included in the EMC test report. Measurements shall be performed from the equipment case to the shell of case mounted connectors J2, J3, J4, J5, J6, J7, J9 and all other externally mounted connectors. Additionally, measurements shall be performed between all mating surfaces bonded by EMI gasketing. (See 3.20.)

4.20 Time totalizing meter. The operational check of paragraph 3.7.12, MIL-M-7793, shall be performed at standard environmental conditions and at nominal ac voltage to determine that the time totalizing meter registers correct time. (See 3.5.)

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4.21 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.21. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, where such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.22 Burn-In. All AN/AYA-10( ) systems shall be operated for a minimum period of 65 hours at the standard environmental conditions of 4.7.1. Operation shall consist of applying the simulated data input of 4.11 and observing the corresponding output display on the monitoring equipment. A failure is defined as the inability of the equipment to meet the requirements of 3.17.1 and 3.17.2 during burn-in and Groups A and B inspection after burn-in. If a failure occurs, a failure analysis shall be performed, the failure shall be repaired, and the AN/AYA-10( ) system shall continue operation until all the following conditions have been met. (See 3.14.)

a. A minimum of 65 hours of burn-in time for the AN/AYA-10( ) system.

b. A minimum of 10 hours of burn-in time without failure for each system individual part, i.e., transistor, IC flat pack, diode, resistor, etc., which has been replaced to correct a failure.

c. A minimum of 30 hours of burn-in time without failure for each system subassembly, i.e., microbeam assembly, printed circuit card assembly, high voltage module assembly, etc., which has been replaced to correct a failure.

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4.23 Reliability evaluation tests. The contractor shall conduct two (2) 800 hour reliability evaluation tests. The initial test shall be performed on the first 4 production samples. The second test shall be performed on production samples immediately following the first thirty five (35) AN/AYA-10( ) systems produced. The number of AN/AYA-10( ) systems under test, for each test, shall be four (4). The 800 hour test duration shall consist of 200 hours of equipment operating time ("on" time) on each system under test. The 200 hours of operating time on each system do not have to be continuous. (See 3.13.)

4.23.1 Reliability test conditions. The reliability test shall be conducted in accordance with test Level E of MIL-STD-781. Operation shall consist of applying the simulated data input of 4.11 and observing the corresponding output display on the monitoring equipment. A failure is defined as the inability of the equipment to meet the required performance parameters as specified in 3.17.1 and 3.17.2. If a failure occurs, a failure analysis shall be performed and the failure shall be repaired. Operating time accumulated between the recognition of a failure and the verification of repair shall not be counted toward the required "on time" of 200 hours. The following paragraphs of MIL-STD-781 shall apply: 5.1.5, 5.1.6, 5.2.2, 5.4.2.3, 5.4.3, 5.4.4, 5.4.5, 5.4.7, 5.5.2, 5.5.3, 5.6, 5.9 and 5.10. Measurements of the requirements of 3.17.1 and 3.17.2 shall be made at -40°C and at +55°C during the heating period at least once at each of these temperatures every 24 hours of reliability testing.

4.23.2 Reinspection of reliability test samples. Sample units which have been subjected to reliability testing may be accepted on the contract provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A and B inspection.

4.24 Safety engineering evaluation. A safety evaluation shall be conducted to determine that all hazards have been identified and eliminated in order that maximum safety of both operating and maintenance personnel and equipments has been accomplished as specified in the systems safety engineering requirements in 3.15.

4.24.1 Safety inspection. During acceptance inspection a visual inspection shall be performed to determine that all requirements, as a result of the safety engineering evaluation, have been incorporated in the equipment. (See 3.15.)

4.25 Visual and mechanical inspection. The equipment shall be inspected to determine that the workmanship requirements of MIL-E-5400 have been met. Defects shall be classified in accordance with MIL-STD-252. (See 3.22.)

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4.26 CDM CRT display intensity. Rotate the display intensity control throughout its range of operation. The intensity shall vary from extinction to an oversized, defocused dot. (See 3.11.)

4.27 Self test. Simulate at least 8 faults in the system and verify that each fault is indicated by the self test CRT display or the fault indicator light on the CDM or both. Faults shall be immulated in a manner such that there is no resulting system damage. (See 3.10.)

4.28 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified. (See 6.2(b).)

5.1.1 Level A.

5.1.1.1 Cleaning. Airborne Data Annotation System AN/AYA-10( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Airborne Data Annotation System AN/AYA-10( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Signal Data Converter CV-2647( )/AYA-10. Each signal data converter shall be individually packaged method III, as follows: cushion each unit on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

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5.1.1.4.3 Control Display Monitor C-8338( )/AYA-10 or Base, Shock Mount, MT-4539( )/AYA-10. Each control display monitor or base, shock mount shall be individually packaged in accordance with the procedure as specified in 5.1.1.4.2, except that the fiberboard box shall be Grade W6c.

5.1.1.4.4 Consolidation. Place the items, packaged as specified in 5.1.1.4.2 and 5.1.1.4.3, within a close-fitting box conforming to PPP-B-636, Type CF, class weather-resistant. Fill all voids with paperboard conforming to PPP-P-291, Type 1, to prevent movement. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. Each Airborne Data Annotation System AN/AYA-10( ) shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be Level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2(b).)

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Airborne Data Annotation System AN/AYA-10( ), packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, Type IV, Size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support

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the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of Airborne Data Annotation System AN/AYA-10( ), packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Airborne Data Annotation System AN/AYA-10( ); packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of Airborne Data Annotation System AN/AYA-10( ), packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard cap shall be class domestic.

5.2.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Airborne Data Annotation System AN/AYA-10( ), packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The AN/AYA-10( ) Airborne Data Annotation System is used as a means of converting aircraft navigation information and sensor identification into a format suitable for display on recording heads of surveillance sensors. (See 1.1.)

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendments thereto.

b. Levels of preservation and packaging and packing. (See Section 5.)

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. When rough handling and functional tests are required.

e. First Article inspection: Three sample units of each item listed in Section 1 are required for First Article inspection.

f. Place of final inspection.

6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3.)

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example AN/AYA-10W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

Custodian:
ARMY EL

Preparing Activity:
ARMY EL

Project No. 5895-A208

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-A-55369(AL) AIRBORNE DATA ANNOTATION SYSTEM AN/AYA-10( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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